

Design-based test that two means are equal

Presentation Outline

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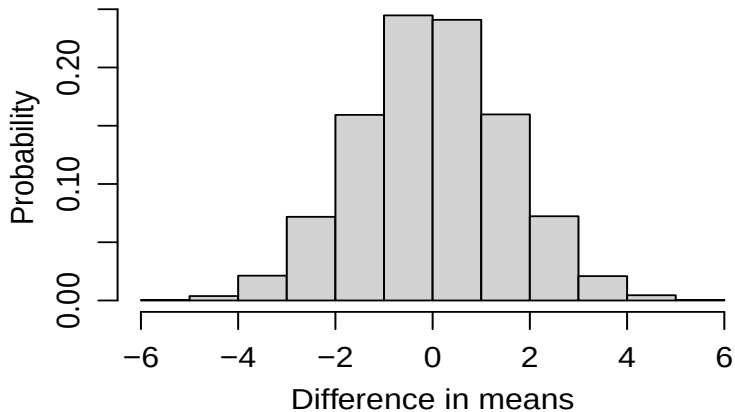
Hypothesis tests

Context:

- Comparing means of two groups
 - Sample means are almost always different
 - Does this mean that the population means are different?
- Randomly draw two samples from the same population
 - Calculate the average for each sample and the difference
 - Repeat many times

Two samples from same population

Histogram of differences between the averages



Logic of a hypothesis test

RA Fisher's proposal

- Identify a null hypothesis
 - Most commonly, absence of the phenomenon of interest
 - Here, that the data in hand are samples from two populations with the same mean
 - Equivalently, the difference in population means = 0
 - Or, the ratio of population means = 1
- Note:
 - The null hypothesis is a statement about the population means
 - Not the sample averages
 - In Fisher's approach, there is no alternative hypothesis
- Identify a test statistic
 - A sample quantity whose distribution depends on whether the null is true or not
 - E.g., difference of two averages, ratio of two averages

Logic of a hypothesis test - 2

Logic:

- Enumerate what sort of data you would see if the null hypothesis were true
- Are the data you have “unusual” in some way
- If so, something is wrong or unlikely
 - null hypothesis is correct and a rare even happened
 - null hypothesis is false (“reject the null”). Usual culprit.
- How to decide how “unusual” are the data?
 - Compare the observed value to the test statistic to what could happen if null is true
 - Calculate the probability of seeing something **as or more extreme** than the observed test statistic

Implementing a hypothesis test

Assume *only* that treatments were randomly assigned to experimental units

- Assume that groups are samples from the same population
- And the sample sizes, n_1 and n_2 , are fixed
- Combine all observations (values) into a single population
 - Randomly choose n_1 values and assign them to treatment 1
 - All other observations (n_2 values) are assigned to treatment 2
 - Calculate sample averages and their difference
 - Called the sampling distribution
 - Calculate $P[\text{random samples are as or more extreme}]$
- Called the p-value for the test

What is more extreme?

When the null hypothesis is equal means, $\mu_A - \mu_B = 0$

- A random difference of 5 is just as extreme as one of -5
- $p = P[|\bar{Y}_A - \bar{Y}_B| \geq \text{random } |\bar{Y}_A - \bar{Y}_B|]$

When the null hypothesis is equal means, $\mu_A/\mu_B = 1$

- A random ratio of 2 is just as extreme as one of $1/2$
- Called a two-sided test or two-sided p-value

One sided test, e.g., null is a negative difference ($\mu_A - \mu_B \leq 0$)

- A random difference of 5 is evidence against the null
- A random difference of -5 is in the null hypothesis space
- $p = P[(\bar{Y}_A - \bar{Y}_B) \geq \text{random } (\bar{Y}_A - \bar{Y}_B)]$

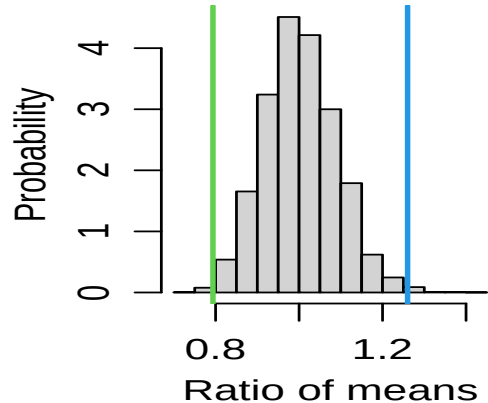
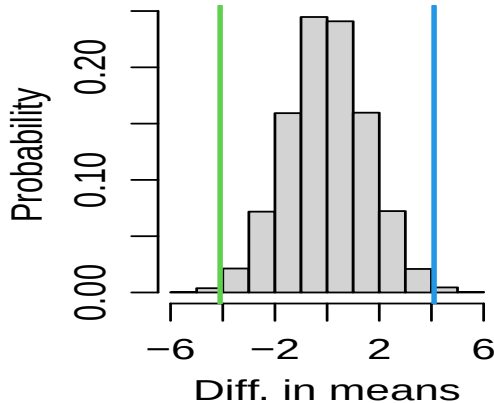
Examples

Summary statistics:

Creativity data			Tomato data		
	n	estimate		n	estimate
Intrinsic	24	19.9	New	7	8.16
Extrinsic	23	15.8	Usual	7	1.63
Difference		4.1			6.53
Ratio		1.26			5.01

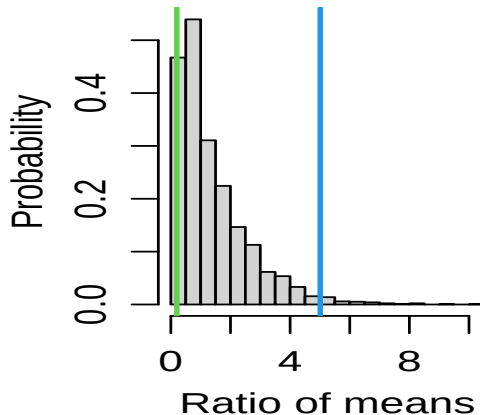
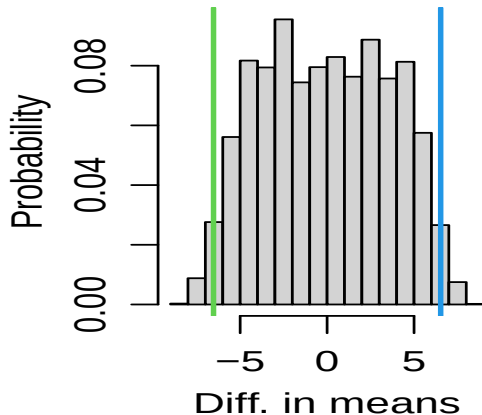
Randomization distributions: Creativity

two-sided $p = 0.0067$ (for both difference and ratio)



Randomization distributions: Tomato yield

two-sided $p = 0.035$



Important Points

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- Randomization tests only assume treatments are randomly assigned
 - Design-based inference - no model required or used
- Allow very general choice of test statistics
- p value: How extreme is the observed data if we assume the null hypothesis is true
- Test statistic: quantity estimated from the data to decide “how extreme”
- $P[\text{as or more extreme test statistic given } H_0]$